PROJECT SYNOPSIS

Education system of India

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Abstract

- Analyses the Indian school education system using publicly available datasets.
- Compares government and private schools based on infrastructure, enrolment, and teacher availability.
- Identifies educational disparities across states and districts.
- Uses Python for data preprocessing, analysis, and visualization.
- Implements a RAG-based AI agent to answer user queries using educational datasets.
- Provides data-driven insights through interactive dashboards and AI-generated responses.

Introduction

- Education is essential for national development.
- India has government, government-aided, and private schools.
- Educational resources and infrastructure vary across regions.
- Publicly available educational data requires extensive analysis.
- This project combines data analytics and AI to simplify educational data exploration.

Problem Statement

- Educational datasets are distributed across multiple sources.
- Comparing government and private schools requires manual analysis.
- Identifying regional educational disparities is challenging.
- Traditional dashboards offer limited natural-language interaction.
- A unified analytical and AI-powered platform is required.

Solution

“Rag system that analysis the data learns from it and answers to the questions ”

Problem Statement

- Educational datasets are distributed across multiple sources.
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- A unified analytical and AI-powered platform is required.

Objectives

- Collect educational datasets from reliable sources.
- Clean and preprocess collected data.
- Compare government and private schools.
- Analyse enrolment, infrastructure, and teacher availability.
- Identify state-wise and district-wise disparities.
- Develop interactive dashboards.
- Build a RAG-based AI agent for natural-language queries.
- Generate evidence-based responses using retrieved data.

Scope of the Project

- Focuses on school education in India.
- Covers government and private schools.
- Analyses state-wise and district-wise data.
- Examines infrastructure, enrolment, and educational resources.
- Supports interactive data visualization.
- Enables natural-language question answering.
- Uses publicly available secondary datasets.

Methodology

1. Data Collection: Gather educational datasets from UDISE+ and other public sources.
2. Data Preprocessing: Handle missing values, duplicates, and inconsistencies.
3. Exploratory Data Analysis: Identify patterns and trends.
4. Comparative Analysis: Compare government and private schools.
5. Visualization: Develop interactive dashboards.
6. RAG Development: Implement document processing, embeddings, and retrieval.
7. AI Integration: Generate answers using retrieved information.
8. Testing: Evaluate analytical accuracy and response quality.

Technologies Used

Component	Technology
Programming	Python
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Plotly
Frontend	HTML, CSS, JavaScript
Backend	Flask
Database	PostgreSQL
RAG Framework	LangChain
Embeddings	Sentence Transformers
Vector Database	FAISS
LLM	Qwen / Gemini
Development	VS Code, Jupyter Notebook

Dataset Requirements

- School identification and management type.
- State and district information.
- Student enrollment statistics.
- Teacher availability.
- Classroom infrastructure.
- Drinking water and sanitation facilities.
- Electricity and internet availability.
- Digital learning resources.

Expected Outcomes

- Consolidated educational dataset.
- Comparative analysis of government and private schools.
- Identification of regional educational disparities.
- Interactive analytical dashboards.
- RAG-based conversational AI agent.
- Evidence-supported answers to user queries.
- Accessible educational insights for researchers and stakeholders.

Applications

- Educational research and analysis.
- Government policy research.
- School infrastructure assessment.
- Educational data exploration.
- Academic projects and studies.
- Public awareness of educational disparities.

Limitations

- Depends on publicly available data.
- Missing or inconsistent records may affect analysis.
- Dataset reporting years may differ.
- AI responses depend on retrieved information quality.
- Cannot independently measure teaching quality or establish causation.

References

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